# CE205 Databases and Information Retrieval

# Lab 4

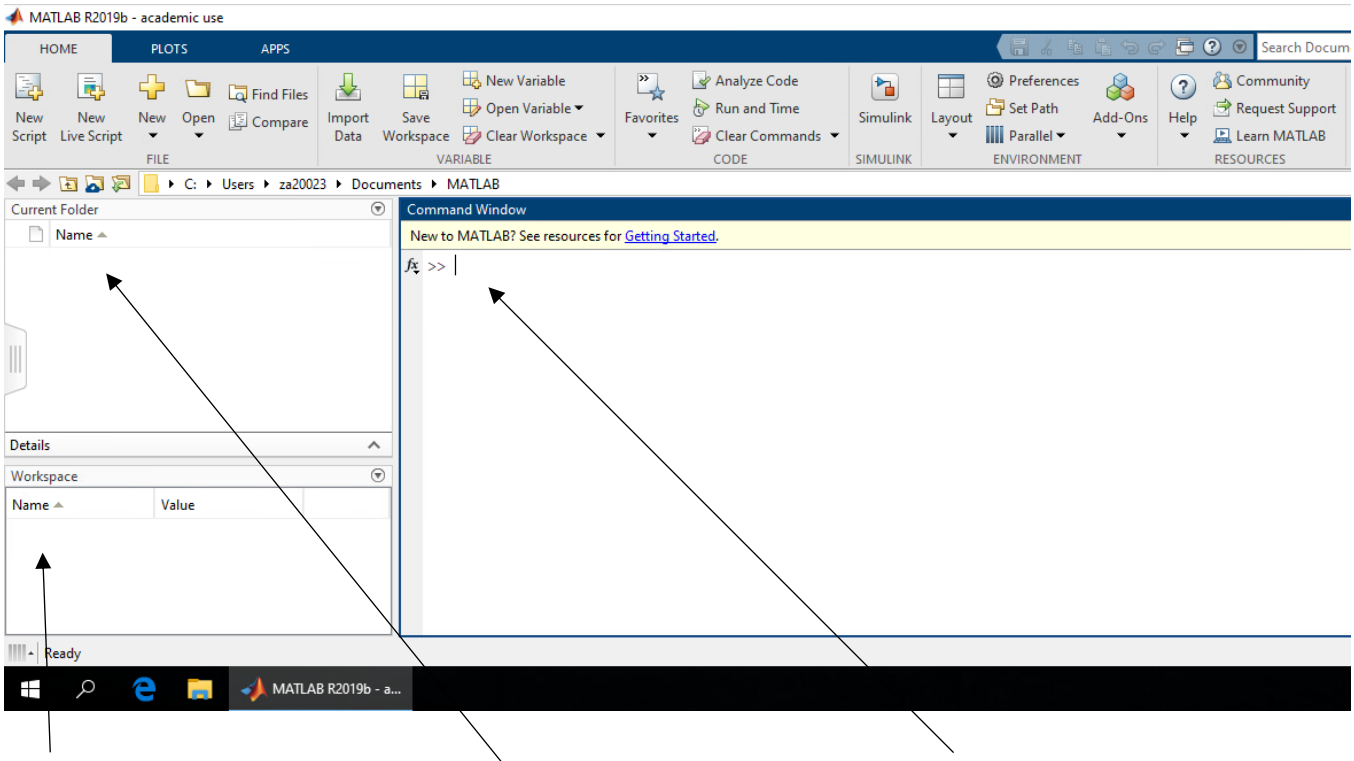
# Design of a Classifier using MATLAB

## Objective:

**The purpose of this activity is to implement a supervised machine learning classification system for discrimination of two classes.**

## MATLAB Interface

1. Open MATLAB
2. The main working window in MATLAB is called the desktop. When MATLAB is started, the desktop appears in its default layout.

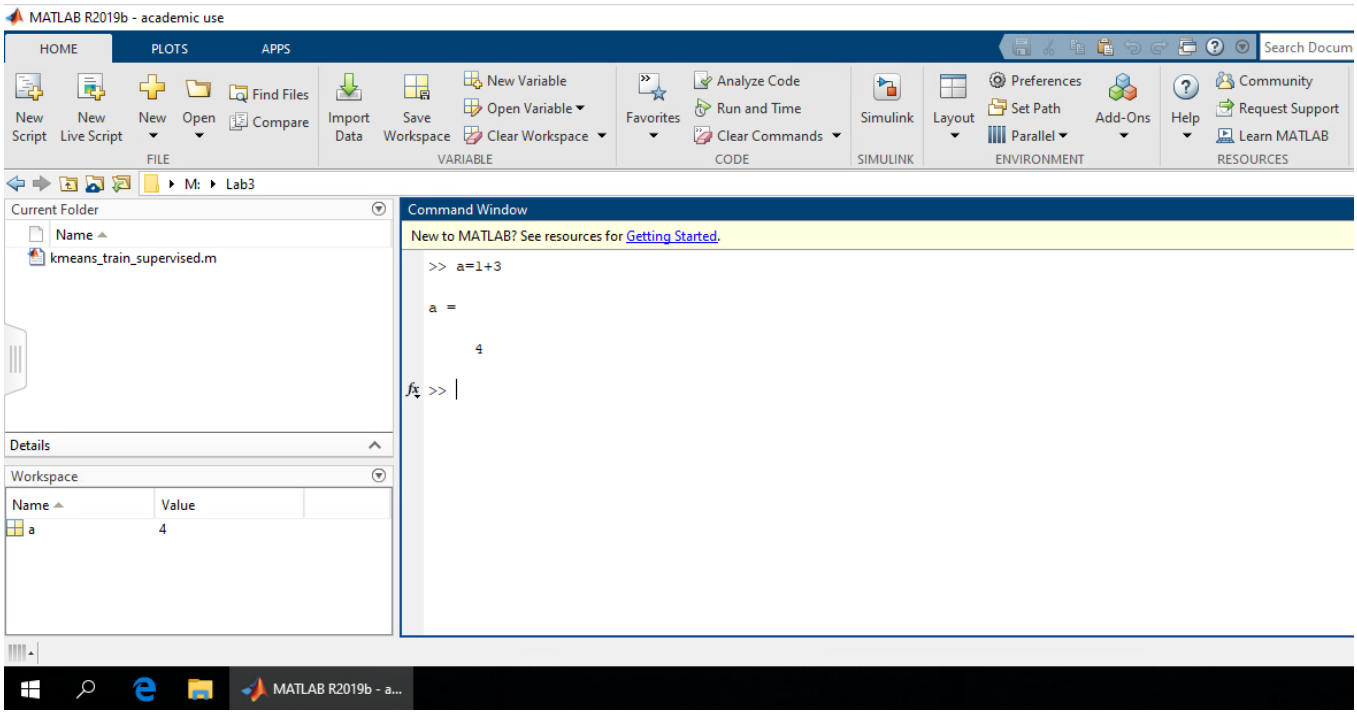



**Workspace** − The workspace shows all the variables created and/or imported from files.

**Current Folder** − This panel allows you to access the project folders and files.

**Command Window** − This is the main area where commands can be entered at the command line. It is indicated by the command prompt (>>).

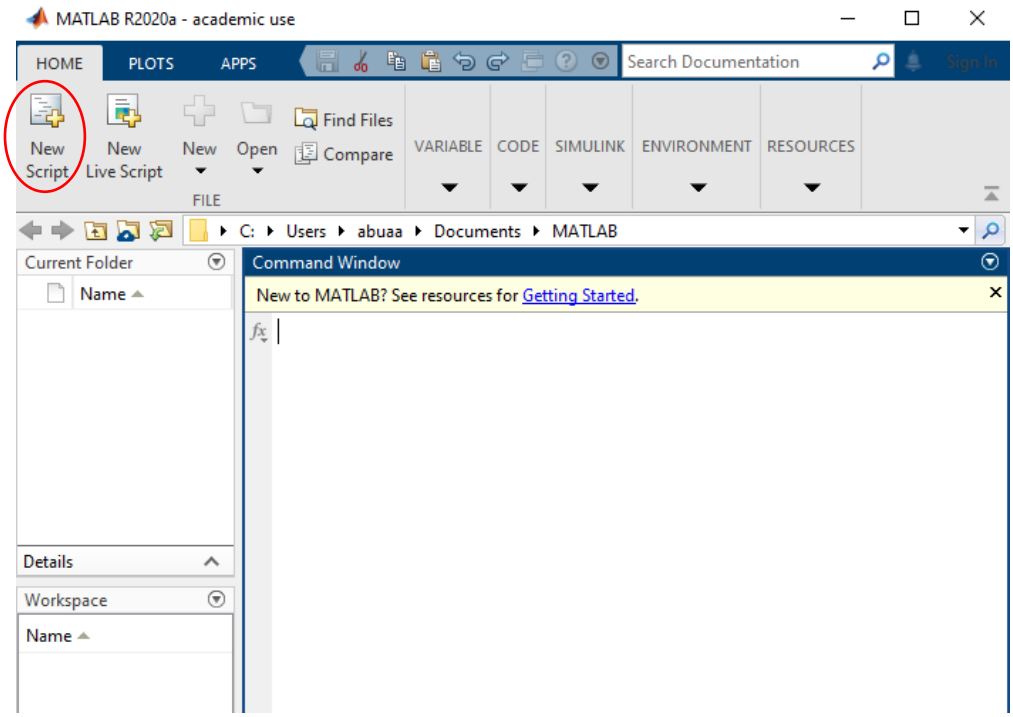
3. If you will execute any statement in the command window, that variable will appear in the workspace.




4. Each variable is stored in form of a matrix. This is the reason MATLAB is called 'Matrix Laboratory'.

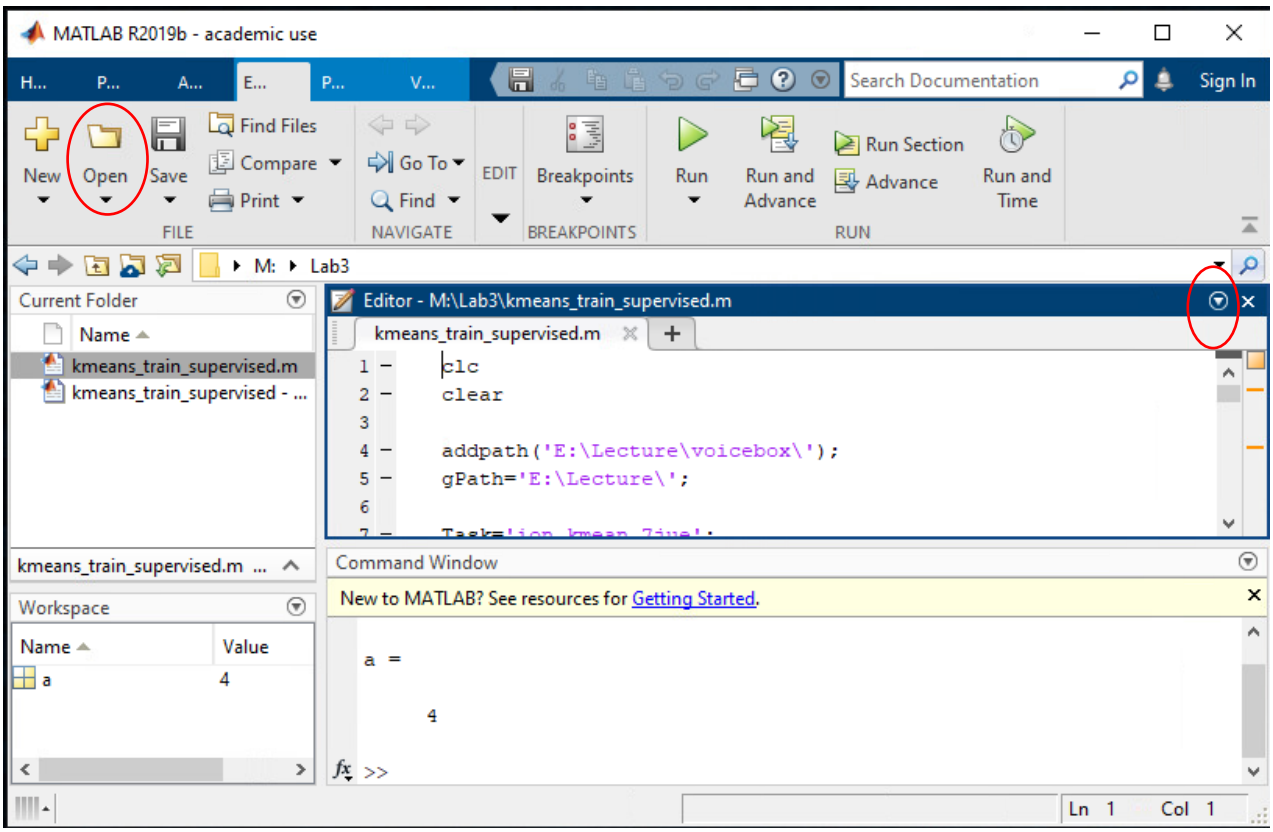
## Create a New File in MATLAB

5. Click on **New Script** to create a new file to write the code.

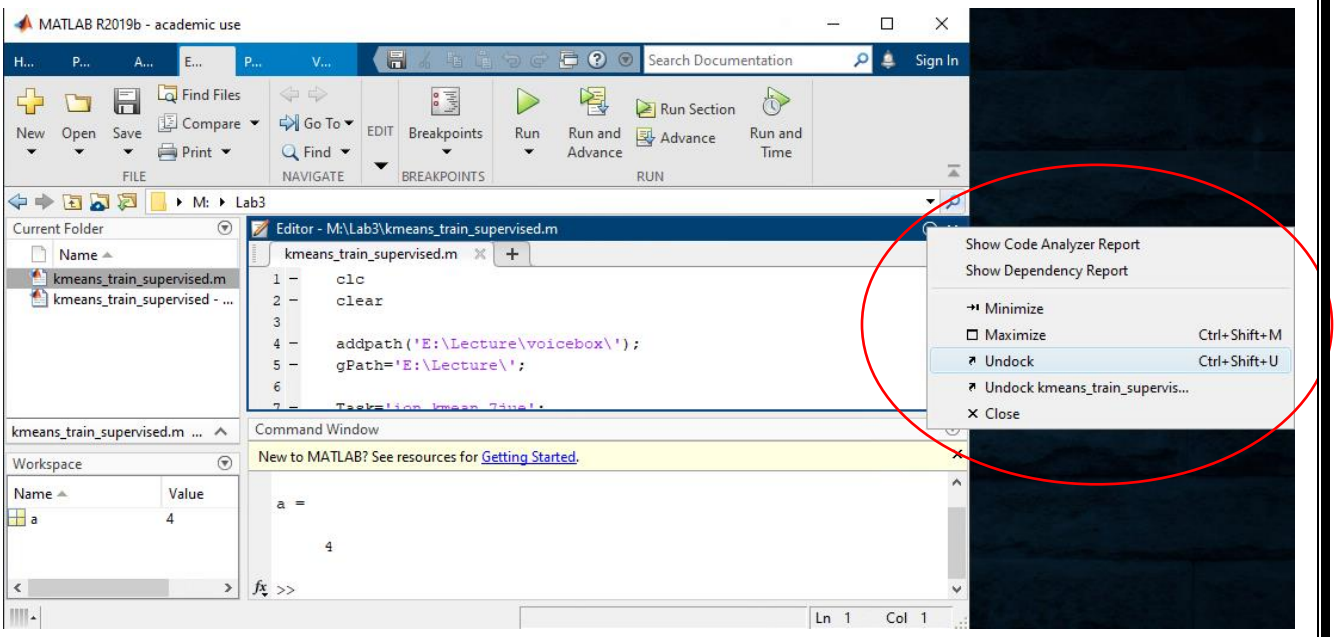


## Working with Existing File

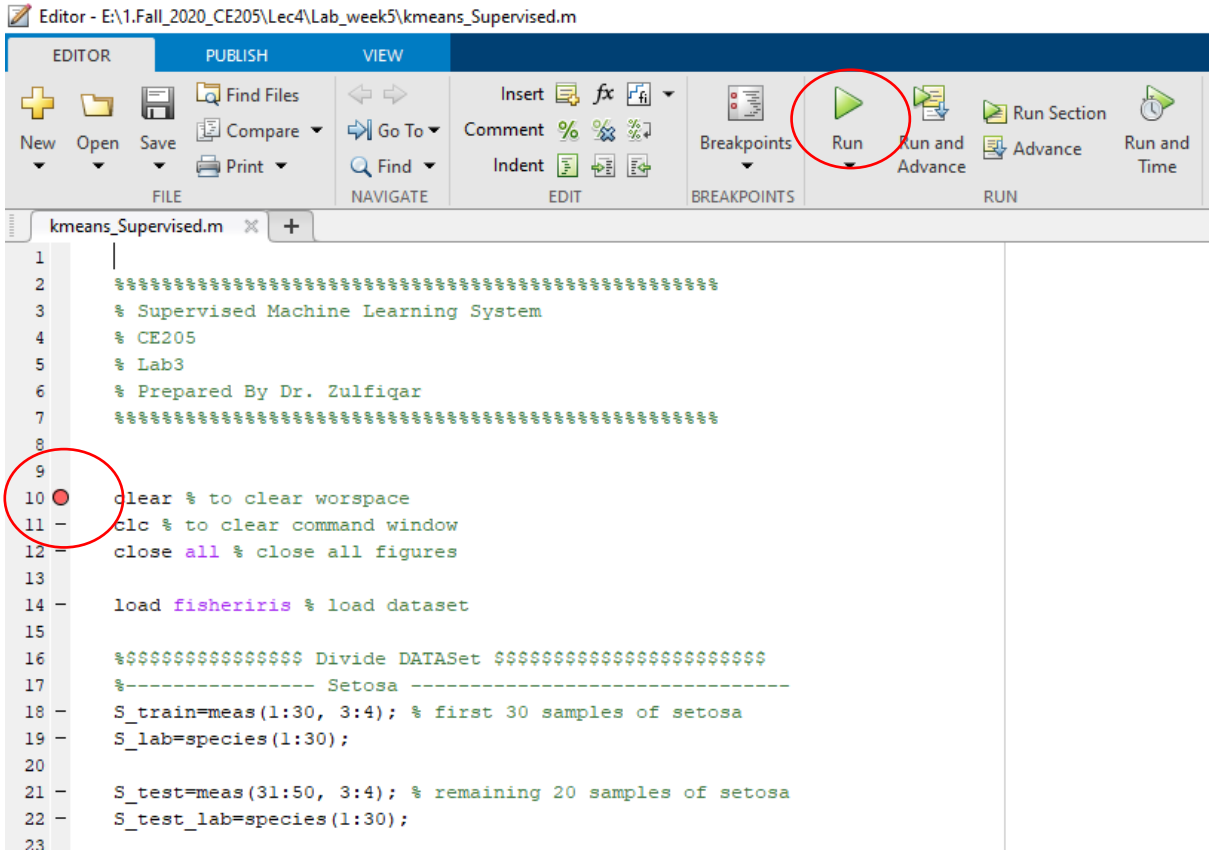
6. To open a file in MATLAB, click on **Open**. You will get the following screen.




7. Undock the file into a new window




8. The file will open in a new window. Now, set a breakpoint by clicking the dash '-' at line number 10 and run the file.


Editor - E:\1.Fall_2020_CE205\Lec4\Lab_week5\kmeans_Supervised.m
EDITOR
PUBLISH
VIEW
New
Open
Save
Find Files
Compare
Print
FILE
Go To
Find
NAVIGATE
Insert
Comment
Indent
EDIT
Breakpoints
BREAKPOINTS
Run
Run and Advance
Run Section
Advance
Run and Time
RUN
kmeans_Supervised.m
1 %%%%

## Resources

Fisher's Iris Data

https://ch.mathworks.com/help/stats/classification-example.html;jsessionid=c324a66c6a22c12b2c20e9a73f51

k-means clustering

https://uk.mathworks.com/help/stats/kmeans.html

# Instructions to Design a Classifier using *k-means* Algorithm

## Steps to Design Training Phase

a) Load dataset **fisheriris** (use load command)
   - Two variables will appear in the workspace
     - ***meas*** (150 x 3) contains the measurements of species
     - ***species*** (150 x 1) contains labels for each row of ***meas***

b) Choose any two species (use colon operator for selection), for example
   - setosa (rows from 1 to 50)
   - versicular (rows from 51 to 100)

c) Divide each species into training and testing sets
   - use colon operator to select samples and features such as
     - *variable (row_i : row_j, column_m : column_n)*

d) Choose the number of clusters (like 2 or 3) that you want to generate for each species (class)

e) Generate training model for each species using k-means algorithm with the following command

   *[idxS,CS] = kmeans(trainigData, number_ of_ clusters, 'Options', 'Display','final');*

f) Open variable ***CS*** in the workspace and see what is inside the generated model

**TRAINING IS DONE!**

## Steps to Design Testing Phase

a) Select a test sample

b) Computes its distance from centroid of class1-setosa, for example the distance with the centroid1 can be calculated as

- *DS1 = sqrt( sum( (test_Sample - CS(1,:)) .^ 2 ) );*

c) Similarly, calculate ***DS2*** and ***DS3*** (distances with 2nd and 3rd centroids) and then compute average distance of the test sample from class1 -setosa)

- *mDS=(DS1+DS2+DS3)/3;*

d) In the same way, compute ***DV1, DV2,*** and ***DV3*** (distances of the test sample from the centroids of class2-versicular) and then find the average distance ***mDV***

e) If the average distance of the test sample from centroid1 is shorter than the centroid2, the test sample belongs to class1 otherwise class2

f) Check whether the output label (suggested by the system) and the true label are the same or not

- If yes, it means the prediction of the classifier is true otherwise false

g) Repeat steps *a* to *f* for all test samples and calculate the accuracy of your classifier using the formula

*Accuracy = ( number of truly predicted sample / total samples ) *100*

**TESTING IS DONE!**

## for loop in MATLAB

https://www.educba.com/for-loop-in-matlab/?source=leftnav

```
for index = initial value : end value

    statement

end
```

## Example

```
for a = 1 : 5
    disp(a)
end
```

## If statement in MATLAB

https://www.educba.com/if-statement-in-matlab/?source=leftnav

```
if condition 1

    Statement 1

else

    Statement 2

end
```

## Example

```
x=5;

if x=5
    Y=7;
else
    Y=0
end
```